# Virtual Local Area Networks (VLANs - IEEE 802.1Q)

*Coverage: Broadcast Containment, Port-Based vs Tagged VLANs, IEEE 802.1Q 4-Byte Tag (VID 1-4094), Access vs Trunk Ports, Router-on-a-Stick*

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## 1. IEEE 802.1Q Tagged Framing

- 4-Byte Tag: TPID 0x8100 (16b), PCP (3b QoS), DEI (1b), VID (12b for 4094 VLANs).
- Trunk ports carry tagged frames; Access ports carry untagged frames.

## 2. Inter-VLAN Routing

- Router-on-a-Stick (sub-interfaces) and Layer 3 Switch SVIs.

### ■ KTU University High-Yield Exam Questions

- Q: Explain VLANs, IEEE 802.1Q tagging, and Router-on-a-Stick inter-VLAN routing. (8 Marks)